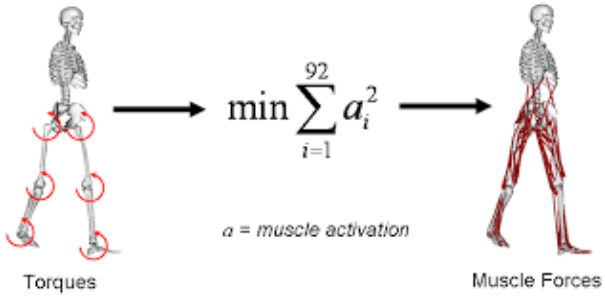
Lecture 13
Optimization in Biomechanics

SM2501/FSG3084

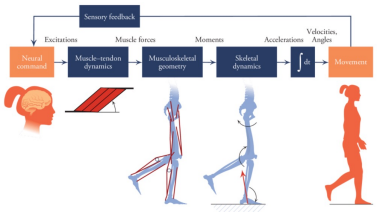


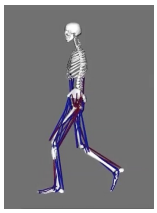
Optimization is useful when there are more unknowns than governing equations.

1

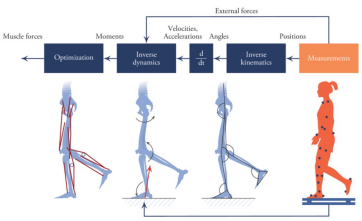
Modelling and measuring neuromuscular control of movement

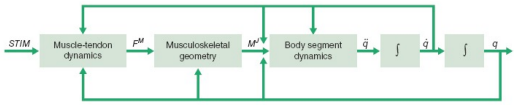
Simulation/Forward Dynamics

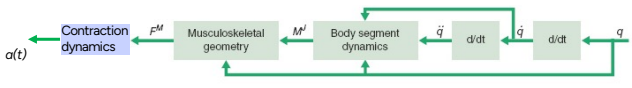




Experimental/Inverse Dynamics







- Inputs: muscle stimulation patterns (STIM)
- Outputs: body segment positions, forces

- Usually ground-up (top-down also possible)
- Inputs: body segment positions (q), external forces
- Outputs: internal reaction forces/moments and muscle forces
- OpenSIM: IK+ID, solving muscle forces with optimization

Uchida and Delp 2020, Robertson et al 2004

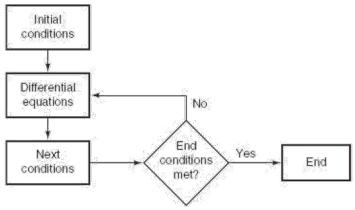
2

Optimization approaches

Goal: to determine which set of solutions will fulfill an **objective criterion**, often to minimize or maximizes a specified **cost function**

Cost function for *tracking kinematics* is usually formulated to minimize the differences between model outputs (e.g. joint angles, GRFs) and corresponding experimental measures. OpenSIM IK, for example, minimizes the error between measured and modelled markers.

Cost functions for e.g. *resolving muscle redundancy* can be related to efficiency (ex. minimize muscular effort) or to an aspect of the motion (e.g. maximize jump height), or even combinations.



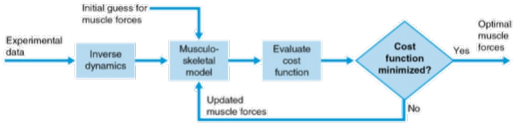
General flowchart of an iterative solution process

3

Optimization approaches

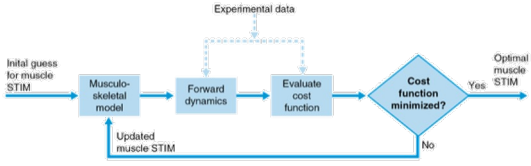
Static Optimization

- The cost function is evaluated at each time step during a movement, independent of any prior or subsequent time steps



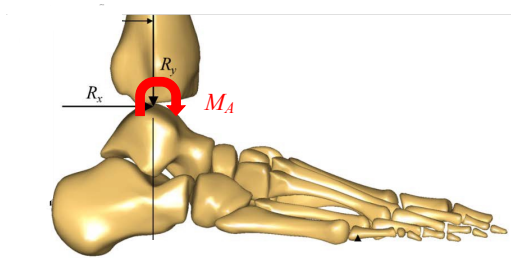
Dynamic Optimization

- The entire movement sequence must be simulated to determine the value of the cost function



4

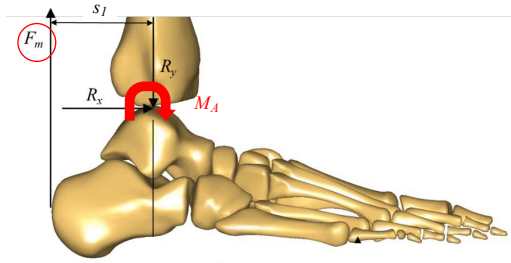
Solving muscle load sharing with static optimization (SO)



- Newton-Euler equations describe dynamic equilibrium.
- In 2D, 3 equations, 3 unknowns
  - Determinate system, reaction forces and moment (R_x, R_y, M_z) can be solved
- In 3D: 6 equations, 6 unknowns
  - Determinate system, reaction forces and moment (R_x, R_y, R_z, M_x, M_y, M_z) can be solved
- (Reaction **forces**: not to be confused with joint contact forces, no direct physiological meaning)
- Reaction **moments**: describes the net requirement of the muscles (and other tissues) to perform this movement

5

Example: n=1 muscle



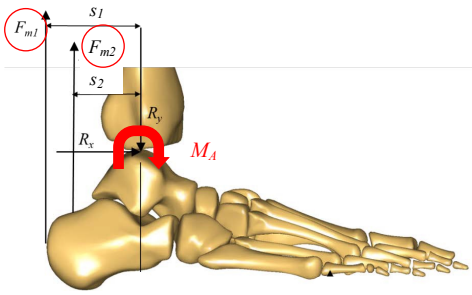
If M_A known and attributed to n=1 muscle with known moment arm s_1 then F_m can easily be solved.

After F_m is estimated, a_m can then be solved via contraction dynamics.

But this is practically never the case.

6

Example: n=2 muscles



M_A is assumed to be the result of F_{m1} and F_{m2}

$s_1F_{m1} + s_2F_{m2} = M_A$

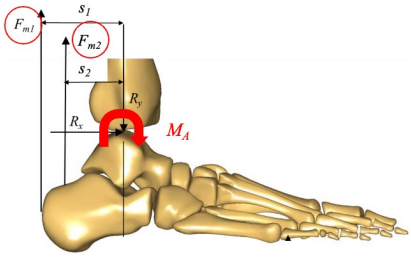
One equation, 2 unknowns....

->There are many possible solutions. This is known as the *muscle redundancy or muscle load-sharing problem*. It is often approached with optimization to find the set of muscle forces that is optimal according to some criterion.

7

Equality constraint around ankle joint:

$s_1F_{m1} + s_2F_{m2} = M_A$



Strategies and solutions:

- Minimize sum of forces -> muscle with largest moment arm allotted all force

  $F_{m2}=0, \quad F_{m1} = M_A/s_1$

  (unlikely, as muscles usually act in synergy)

- Equal muscle forces

  $F_{m1} = F_{m2} = M_A/(s_1+s_2)$

  (unlikely, as they probably have different strengths)

And so on...

- A common approach is to scale to muscle size/PCSA, which encourages *load sharing*, such that *activation* in each muscle is as evenly distributed as possible. This accounts for both moment arms and muscle strengths/PCSA.

8

General Optimization Problem

$$
\begin{aligned}
&\text{minimize} && J(\underline{x}) && \left.\begin{array}{l}\text{Adjust design variables } \underline{x}\\ \text{to minimize objective}\\ \text{function } J(\underline{x})\end{array}\right\}\\
&\text{subject to} && g_i(\underline{x}) \le 0, \quad i = 1, \ldots, n^i && \left.\begin{array}{l}\text{while satisfying } n^i\\ \text{inequality constraints,}\end{array}\right\}\\
& && h_j(\underline{x}) = 0, \quad j = 1, \ldots, n^j && \left.\begin{array}{l}\text{satisfying } n^j \text{ equality}\\ \text{constraints,}\end{array}\right\}\\
& && \underline{x}^{\text{lower}} \le \underline{x} \le \underline{x}^{\text{upper}} && \left.\begin{array}{l}\text{and respecting bounds}\\ \text{on the design variables.}\end{array}\right\}
\end{aligned}
$$

Design variables (e.g. muscle activations) to be solved.
Objective or cost function: an expression that quantifies the favorability of a particular set of design variable values, for instance to be minimized or maximized.
Constraints: expressions (equalities or inequalities) that must be satisfied for the solution to be feasible.

If there are too many constraints, the solution space may be empty; it may be acceptable to convert one or more strift constraints into soft constraints, or added “penalties” to the objective fuction which can be given weights.

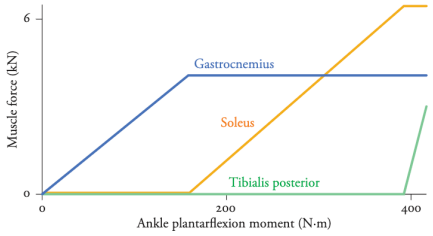
Uchida & Delp 2020

9

Example

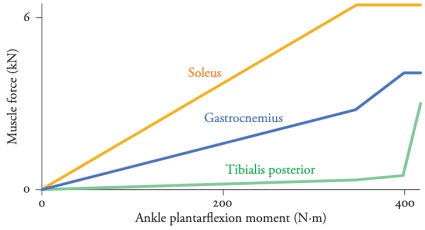


Solution for **minimal sum of muscle force** for $M_A \le 400$ N·m:



(Unrealistic)

Solution for **minimal muscle stresses** for $M_A \le 400$ N·m:



(Matches observed behavior better)

Uchida & Delp 2020

10

Common objective functions in SO

Minimum muscle activations to some power p

$f = \sum_{i=1}^{n} a_i^p$ $\quad i = 1, \dots, n$ muscles

Minimum muscle stresses to some power p

$f = \sum_{i=1}^{n} \left(\frac{F_{M,i}}{PCSA_i}\right)^p = \sum_{i=1}^{n} \left(\sigma_{M,i}\right)^p \quad i = 1, \dots, n$ muscles

p typically 2,3,4, with increasing load sharing. As $p \to \infty$, this approaches a **min-max criteria**

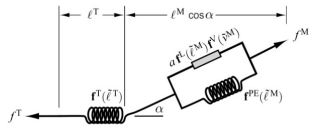
$f = max\left(\frac{F_{M,i}}{PCSA_i}\right) \quad i = 1, \dots, n$ muscles

11

Contraction dynamics

Muscle force (f^M) commonly computed as

$$f^M = f_0^M(a\mathbf{f}^{\mathrm{L}}(\tilde{l}^M)\mathbf{f}^{\mathbf{V}}(\tilde{v}^M) + \mathbf{f}^{\mathrm{PE}}(\tilde{l}^M))$$

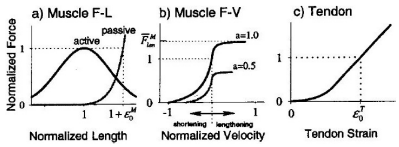


a: muscle activation [a_{min}, 1]

'Hill-type' muscles. Can be modified for more or less detail

- f_0^M : peak force at a length of l_0^M
- $\mathbf{f}^{\mathrm{L}}(\tilde{l}^M)$: active force length curve (length normalized by l_0^M)
- $\mathbf{f}^{\mathbf{V}}(\tilde{v}^M)$: force-velocity curve (velocity normalized by v_{max}^M)
- $\mathbf{f}^{\mathrm{PE}}(\tilde{l}^M)$: passive force-length curve (length normalized by l_S^T)

(Millard et al 2013)



Tendons often modelled as **noncompliant** in static optimization

To solve for muscle *excitation*, first-order **activation dynamics** can be applied

- Represents rate of change of muscle *activation* (concentration of calcium ions) to the muscle *excitation* (firing of motor units) (Thelen et al 2003)

12

In OpenSIM

Motion is entirely determined by generalized coordinates (positions, velocities and accelerations). The SO tool uses the *known* motion of the model to solve the equations of motion for the *unknown* generalized forces, constrained by muscle parameters for force-length-velocity behavior

$$\sum_{m=1}^{n} [a_m f(F_m^0, l_m, v_m)] r_{m,j} = \tau_j$$

While minimizing the cost function:

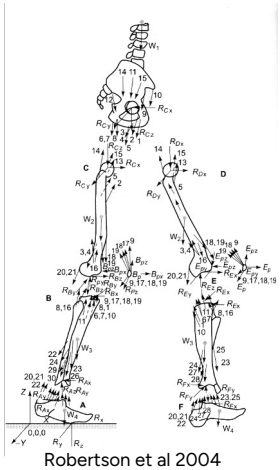
$$J = \sum_{m=1}^{n} (a_m)^p$$

Where n=number of muscles, a_m is activation level of muscle m at a discrete time step, F_m^0 is the maximum isometric force of muscle m, v_m is its shortening velocity, $f(F_m^0, l_m, v_m)$ is the force-length-velocity surface, $r_{m,j}$ is its moment arm around joint j, and p is a user-defined exponent.

Note that for static optimization, $f(F_m^0, l_m, v_m)$ refers to the active fiber force along the MTU, assuming an inextensible tendon and excluding the muscle's parallel elastic element. Activation dynamics are also commonly ignored.

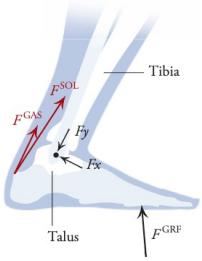
13

Joint contact forces



Robertson et al 2004

- Sum of all forces from muscles, ligaments
- Can only be estimated after resolving muscle load sharing problem



Uchida & Delp 2020

14

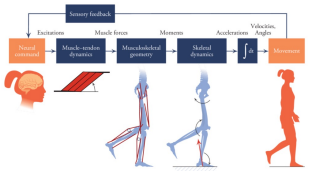
Does static optimization imply static equilibrium?

- No. Static optimization solves for each instant in time independently. Can be considered an extension of inverse dynamics.
- Advantages: low computational cost
- Disadvantages: time-dependent physiological nature of muscles is not considered

15

Dynamic optimization (DO)

Applied in conjuction with forward dynamics models, which simulate the motion of the body from a given set of joint moments (torque-driven) or muscle forces (muscle-driven).



Generalized equations of motion:

$$\ddot{\boldsymbol{q}} = M^{-1}(\boldsymbol{q})\{\boldsymbol{F^G}(\boldsymbol{q}) + \boldsymbol{F^C}(\boldsymbol{q}, \dot{\boldsymbol{q}}) + R(\boldsymbol{q})\boldsymbol{F^T}(\boldsymbol{u}) + \boldsymbol{F^E}(\boldsymbol{q}, \dot{\boldsymbol{q}})\}$$

$q, \dot{q}, \ddot{q}$: positions, velocites and accelerations of the joints
$M^{-1}(\boldsymbol{q})$: inverse of the system mass matrix (masses & inertial properties of segments)
$\boldsymbol{F^G}(\boldsymbol{q})$: forces arising from gravity
$\boldsymbol{F^C}(\boldsymbol{q}, \dot{\boldsymbol{q}})$: Coriolis and centrifugal forces (for laws of motion applied to rotating coordinate systems)
$R(\boldsymbol{q})$: muscle moment arms
$\boldsymbol{F^T}(\boldsymbol{u})$: tendon (MTU) forces as function of excitations u
$\boldsymbol{F^E}(\boldsymbol{q}, \dot{\boldsymbol{q}})$: external forces applied to the model, including GRFs.

DO models often designed to find muscle stimulation patterns that result in an optimal motion wrt the performance criteria.

16

Fundamental and practical differences between SO and DO

- In SO, kinematics are *constrained*, for instance by experimental data
- In DO, kinematics are *computed*

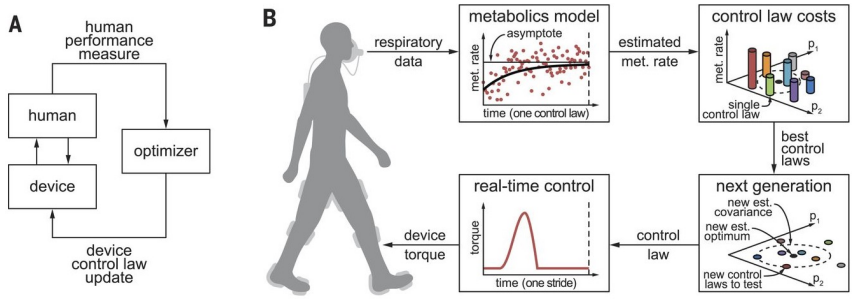
- SO is not static; states can be dynamic. Each time frame is considered independently

- In DO, time to perform a motion influences the solution

- SO is comparatively computationally efficient for many muscles
- DO models have often been 2D stick figures, torque-driven or only with a few muscles

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Human-in-the-loop optimization
(ex. exoskeleton parameter design)



Performance (here, oxygen cost) monitored, design parameters are modified (torque, timing, etc), influence on subject used as feedback in iterative process, e.g. measured metabolic consumption, muscle/muscle group force, joint contact force, etc., even subjective, e.g. ”comfort”.

Zhang et al. 2017, Science

18

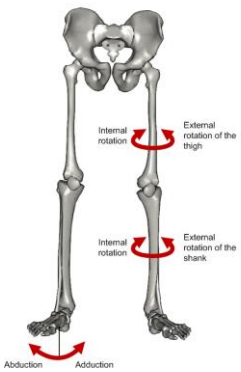
Shoulder modeling
Project 3

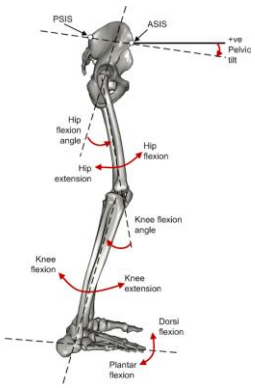
Jennifer Maier

1

Lower body

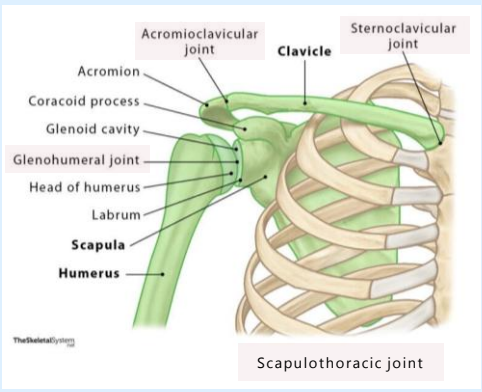
Walking: Mostly in sagittal plane





Upper body

Shoulder enables high range of 3D motion

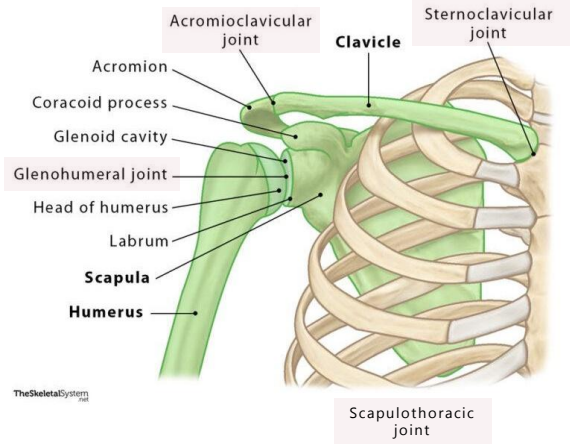


2

Image: https://www.theskeletalsystem.net

2

Shoulder joint anatomy



Role of the scapula

- Stable base for the humeral head
- Retraction and protraction along thoracic wall
- Elevation of acromion to decrease impingement
- Important link in the kinetic chain

[Kibler et al., *Am J Sports Medicine*, 1998]
[Paine et al., *Int J Sports Physical Therapy*, 2013]
Image: https://www.theskeletalsystem.net

3

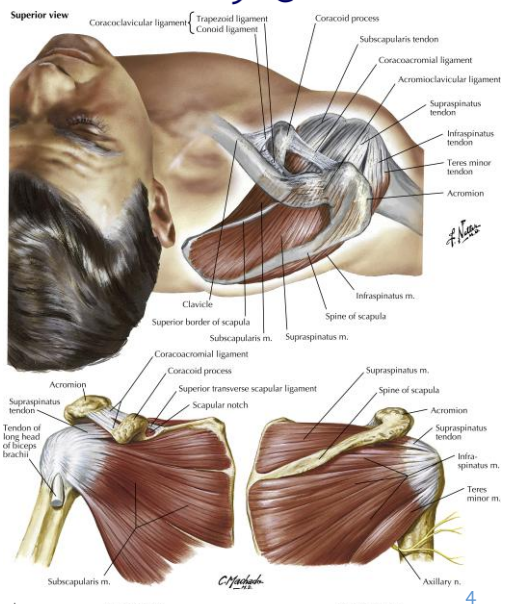
3

Shoulder muscles & function (selection)



Rotator cuff muscles

- Stabilize glenohumeral joint (all arise from scapula and insert into humerus)
- *Subscapularis*: internal shoulder rotation
- *Supraspinatus*: shoulder abduction 0 – 15 degrees
- *Infraspinatus*: external shoulder rotation
- *Teres* Minor: external shoulder rotation



Netter Anatomy

4

4

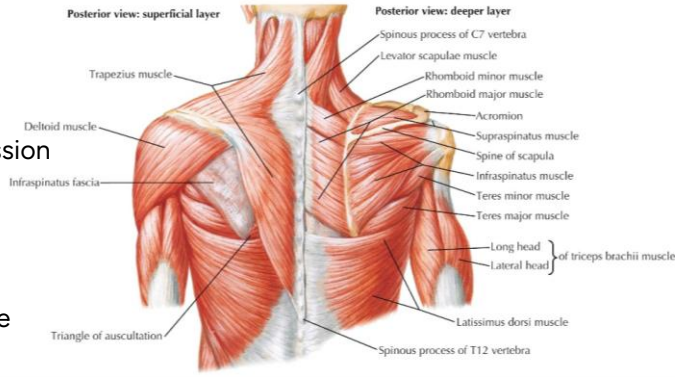
Shoulder muscles & function (selection)

Trapezius

- Upper: scapula upward rotation
- Middle: scapula retraction
- Lower: scapula upward rotation and depression

Rhomboids

- Stabilize medial border of the scapula
- Scapula adduction/retraction towards spine



Netter Anatomy

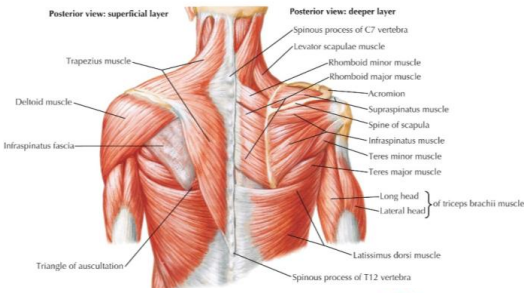
Shoulder muscles & function (selection)

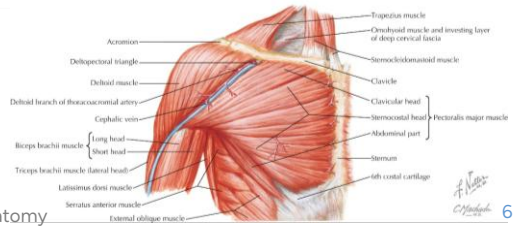
Deltoid

Elevate arm from 15 – 100 degrees

Serratus anterior

- Stabilizes scapula during elevation
- Pull scapula forward and around on thoracic cage ("boxing muscle")

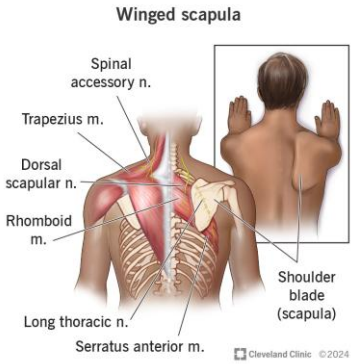




Netter Anatomy

Shoulder dysfunction

Scapular dyskinesis



Causes
Injuries to the nerves or muscles that support affected scapula

Symptoms
- Shoulder pain
- Stiffness
- Reduced range of motion
- Weakness

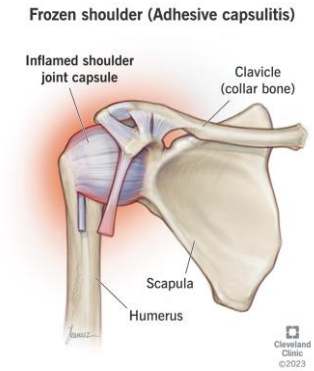
Treatment
- Pain management
- Physical therapy

7

7

Shoulder dysfunction

Adhesive Capsulitis



Causes
Not definite. Occurs when inflammation causes your shoulder joint capsule to thicken and tighten

Symptoms
- Shoulder becomes stiff and is painful to move
- May worsen at night

Treatment
- Medication to reduce pain and swelling
- Hot and cold compress
- Physical therapy

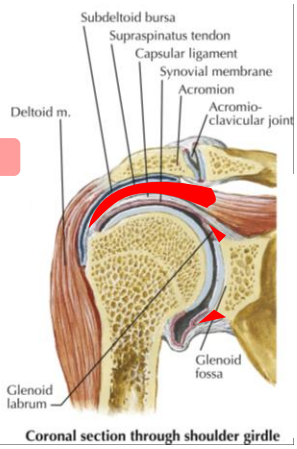
8

8

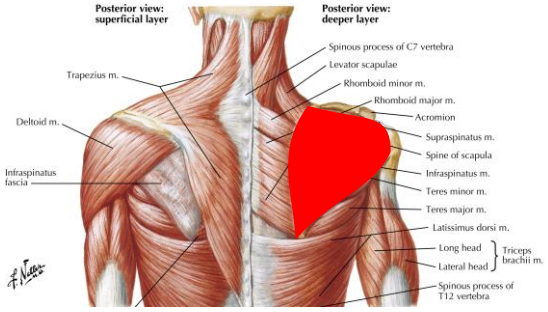
Shoulder (sport) injuries

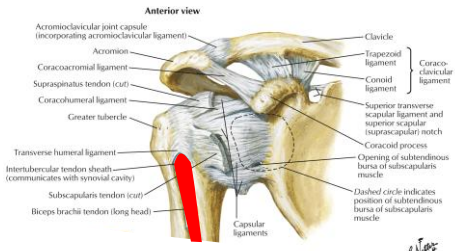
- Labrum injuries
- Tendinitis
- Impingement syndrome
- Rotator cuff tears





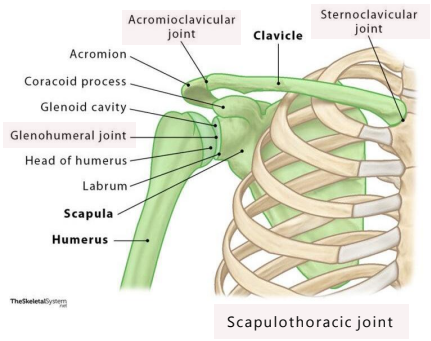
Coronal section through shoulder girdle





Netter Anatomy

Build your kinematic shoulder model



Scapulothoracic joint

What types of joints?

How many degrees of freedom?

Which range of motion?

WeldJoint
introduces no degrees of freedom and fuses bodies together

PinJoint
one coordinate about the common Z-axis of parent and child joint frames

SliderJoint
one coordinate along common X-axis of parent and child joint frames

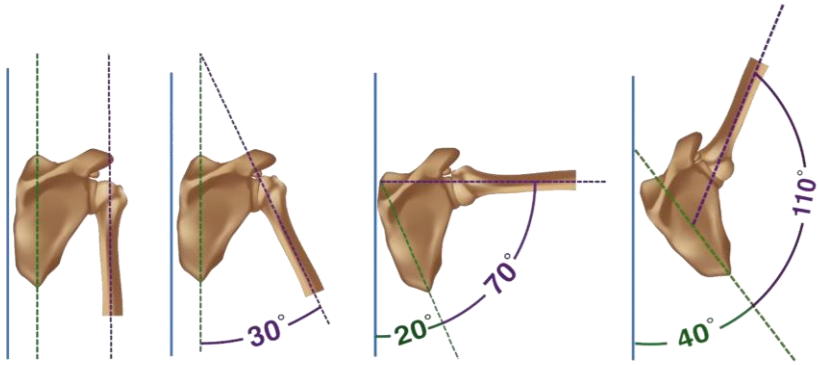
BallJoint
three rotational coordinates that are about X, Y, Z of B in P

CustomJoint
user specified 1-6 coordinates and user defined spatial transform to locate B with respect to P

EllipsoidJoint
three rotational coordinates that are about X, Y, Z of B in P with coupled translations such that B traces and ellipsoid centered at P

Shoulder joint modeling

Scapulohumeral rhythm

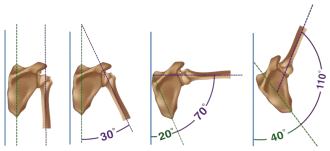


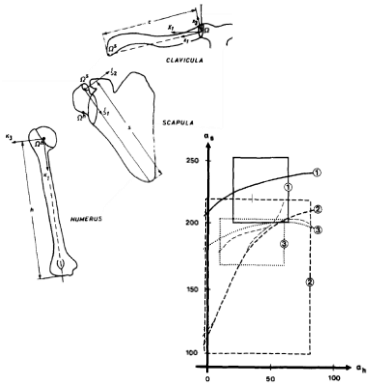
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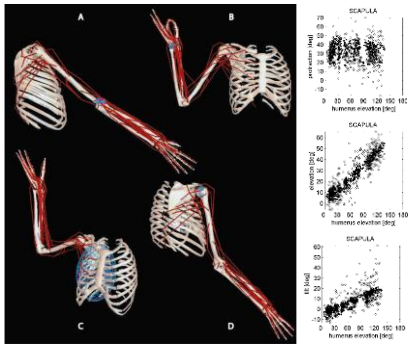
Shoulder joint modeling

Scapulohumeral rhythm





Högfors et al., *J Biomechanics*, 1991



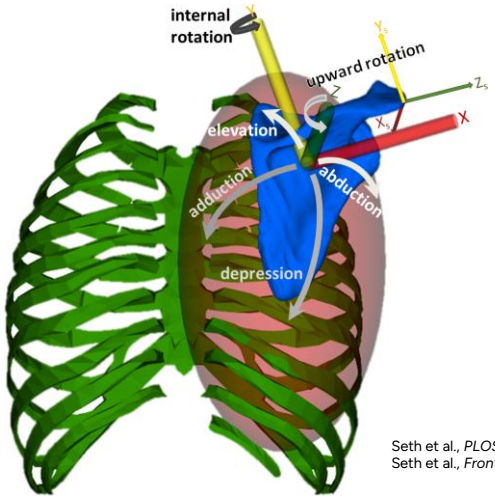
Holzbaur et al., *Annals of Biomed Eng*, 2005
Saul et al., *Comp Methods in Biomech and Biomed Eng*, 2015
De Groot et al., *Clin Biomechanics*, 2001

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Shoulder joint modeling

Scapulothoracic joint

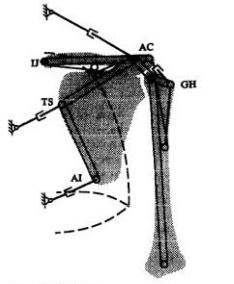


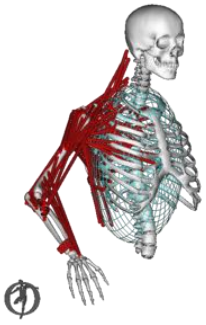
Seth et al., *PLOS ONE, 2016*
Seth et al., *Frontiers in Neurorobotics, 2019*

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Shoulder joint modeling

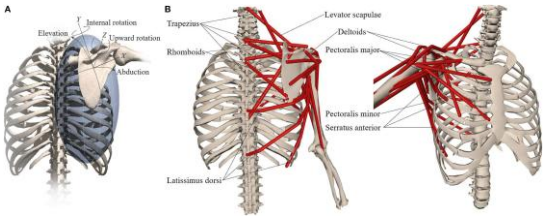




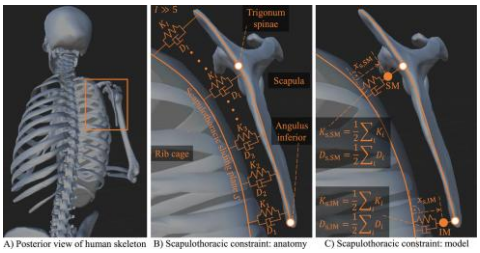
: BEAM (bone)
: HINGE (joint)
: TRUSS (ligament)
: SURFACE (scapulothoracic gliding plane)
: TRUSS or CURVED-TRUSS (muscle)

Blana et al., *J Biomechanics*, 2008
Chadwick et al., *IEEE TBME*, 2014

Van der Helm, *J Biomechanics*, 1994
Happee and van der Helm, *J Biomechanics*, 1995



Seth et al., *PLOS ONE, 2016*
Seth et al., *Frontiers in Neurorobotics, 2019*



A) Posterior view of human skeleton B) Scapulothoracic constraint: anatomy C) Scapulothoracic constraint: model

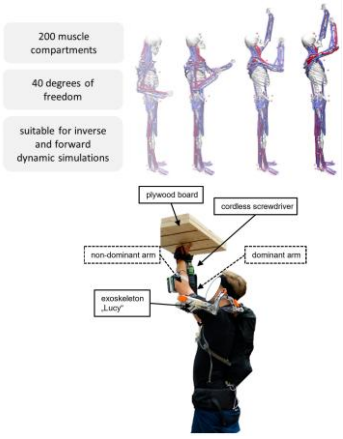
Hu et al., *Comp Methods in Biomech and Biomed Eng*, 2020

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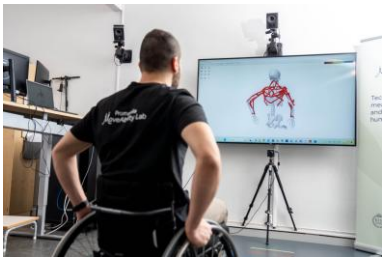
Shoulder research

Occupation/Ergonomics

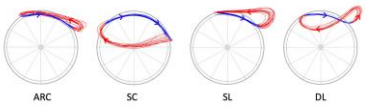


Molz et al, *Appl. Sci*, 2024

Daily life

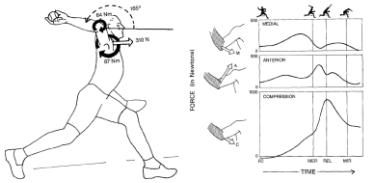


Promobilia MoveAbility Lab



Walford et al., *J Biomech Eng*, 2024

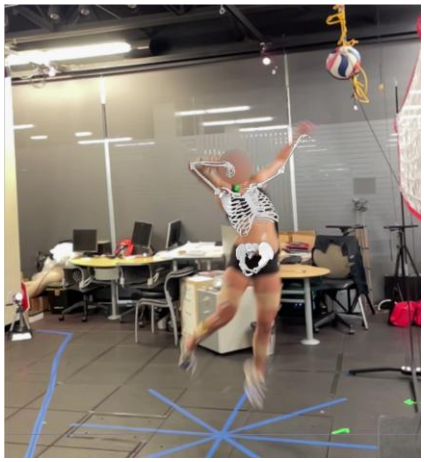
Sport

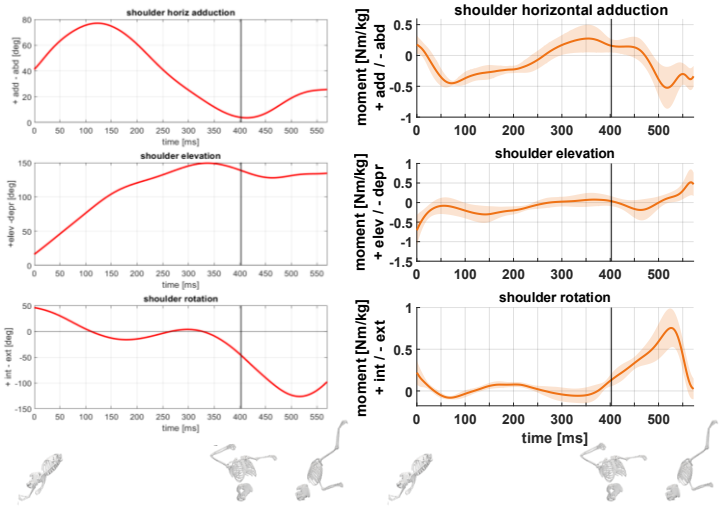


Fleisig 1996, Operative Techniques in Sports Medicine



Volleyball spiking analysis



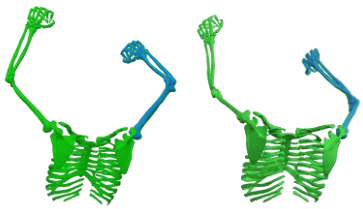


Maier et al, ISB conference, 2025

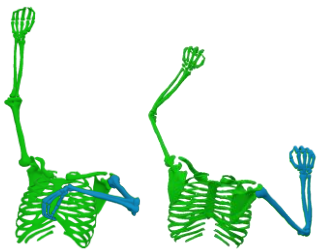
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Volleyball spiking analysis



Coaches



High elbow

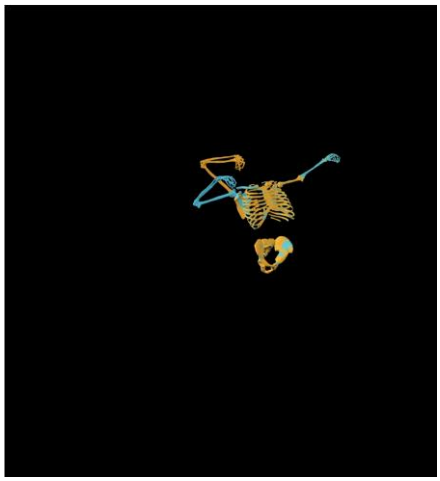
Low elbow

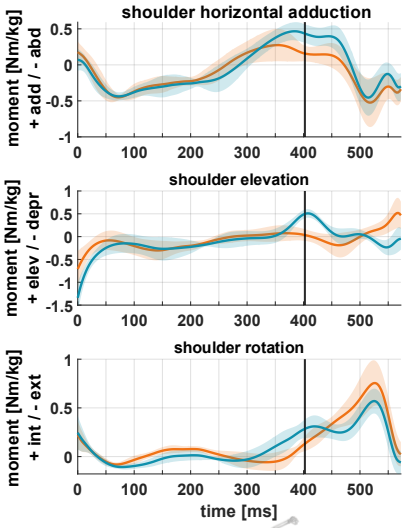
Maier et al, ISB conference, 2025

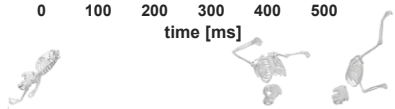
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Volleyball spiking analysis







Maier et al, ISB conference, 2025

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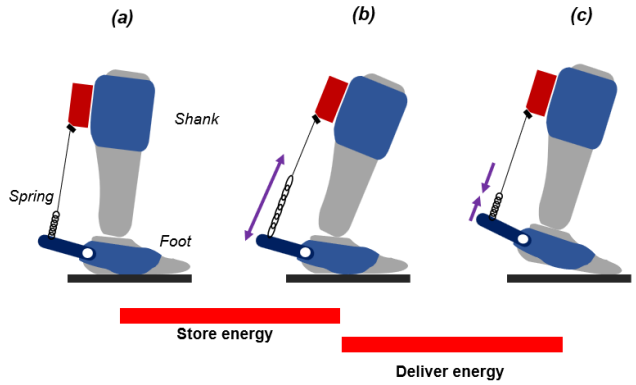
Project 3

Design a passive assistive device for walking

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Passive assistive device



Three parameters

- Resting length
- Stiffness
- Attachment (Moment arm)

Task

Propose the parameters for an extension spring to assist plantarflexion

Goal

Find a spring configuration that minimizes the sum of the activations of the muscles that span the ankle

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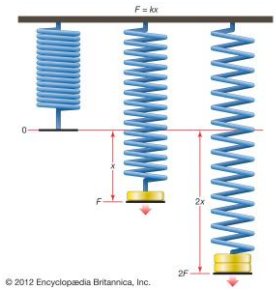
Project info

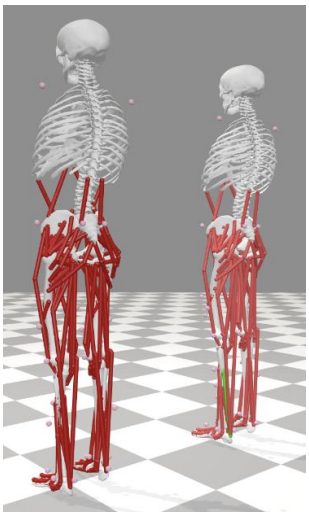
Adding a spring to the model

- Run "addPathSpring.py" to add the spring to OpenSim model
- Run SO with new model given specific spring parameters

Analyze effect of different values for

A. Resting length
B. Stiffness
C. Moment arm





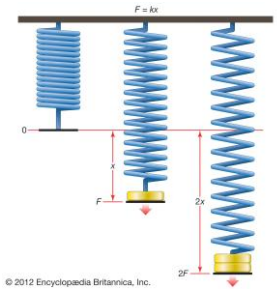
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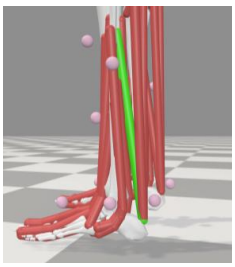
22

Project info

A. Resting length

- Starting values: Resting length = 0.304 m (about the same length as the soleus in a neutral position)
  Stiffness = 120 000 N/m
- Study the effect of shorter or longer resting lengths on the muscle activations in walking using SO
  - Spring force during the gait cycle
  - Activations of muscles that span the ankle
- Find an optimum resting length and motivate why



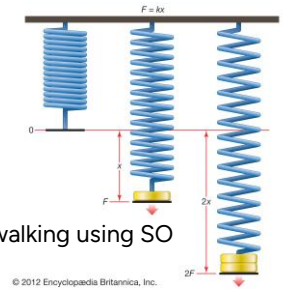


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Project info

B. Stiffness

- Starting values: Resting length = Optimum you found in part A
  Stiffness = 120 000 N/m
- Study the effect of higher or lower stiffness values on the muscle activations in walking using SO
  - Spring force during the gait cycle
  - Activations of muscles that span the ankle
- Find an optimum resting length and motivate why
- Is it possible to completely replace all plantarflexor muscle forces using an extension spring? What would be the implications (advantages/disadvantages)?
- Do the activations of more proximal muscles change with the spring stiffness?

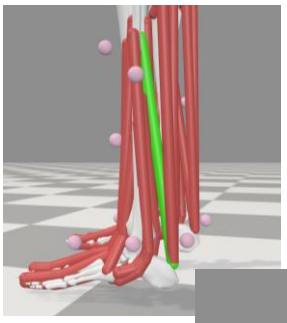


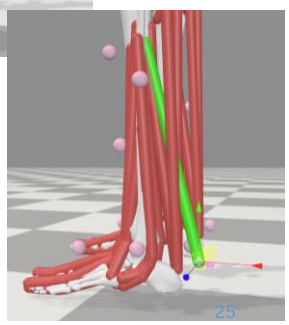
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Project info

C. Moment arm

- Starting values: Resting length = Optimum you found in part A
  Stiffness = 50 000 N/m
- Study the effect of an increased moment arm on the muscle activations in walking using SO. Keep in mind that that the attachment point may affect the spring's length.
  - Spring force during the gait cycle
  - Activations of muscles that span the ankle
- Plot and discuss the contribution of the spring moment to the net ankle plantarflexion/dorsiflexion moment
  -> Use the provided script "calculateSpringMomentArm.py" to compute the moment arm of the spring throughout the motion



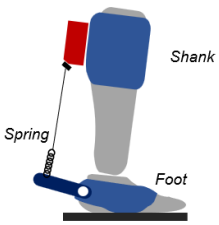


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Project info

D. Design Proposal



- **Propose an assistive device with reasonably realistic characteristics that can reduce muscle activation in walking**
- You are free to modify any or all the spring parameters - resting length, stiffness, and geometric path/attachment points - as long as the spring connects the shank and foot segments.
- Describe the advantages of your design and sketch your proposed assistive device

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Project Deadline **Monday, December 8th at 9:30 am** (before the lecture):

- Write report according the instructions and questions
- Max 8 pages (including text, and figures) and ~4000 words
- Describe the contribution of each team member
- Upload to the Assignments section in Canvas for your group

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